2 **(a)** The gravitational potential ϕ of a point at a distance r from an isolated point mass m is given by the expression

$$\phi = -\frac{Gm}{r}$$

where G is the gravitational constant.

Explain why gravitational potential is negative.

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.... [2]

(b) A space probe of mass m is launched from the geographical North Pole, as shown in Fig. 2.1.

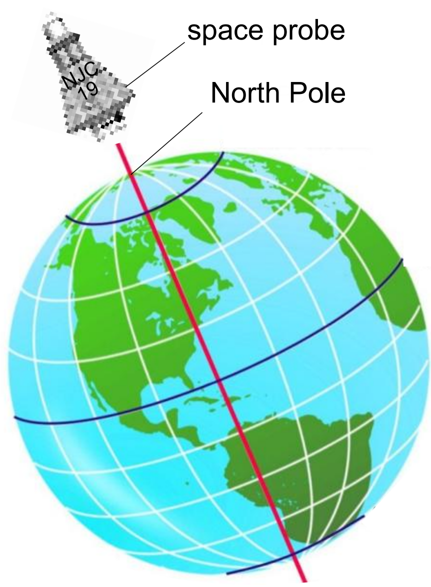

Fig. 2.1

The probe is provided with energy equivalent to an initial kinetic energy E during the launch given by the expression

$$E = \frac{3GMm}{4R_E}$$

where M is the mass of the Earth and R_E is the radius of the Earth.
Work done in overcoming resistive forces in Earth's atmosphere can be neglected.

(i) Explain why the probe cannot reach infinity with this energy E.

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.... [2]

(ii) The mass m of the probe is 1000 kg, the mass M of Earth is 6.0×10^{24} kg and the radius R_E of the Earth is 6.4×10^6 m.

Calculate the distance from the centre of Earth at which the probe will come to a rest momentarily.

distance = ... m [3]

(iii) Explain why the distance calculated in **(b)(ii)** is further if the space probe is launched from the equator in the eastward direction.

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.... [2]

[Total: 9]